

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Ans : HTTP : HyperText Transfer Protocol.

HTTPS : HyperText Transfer Protocol Secure.

FTP : File Transfer Protocol .

SMTP : Simple Mail Transfer Protocol.

2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

Ans : IPv4 uses a 32-bit address and supports about 4.3 billion unique addresses, while IPv6 uses a 128-bit address and supports a much larger number of addresses.

3. Run nslookup google.com on your system and write down the IP address you get.

Ans : google.com

Addresses: 2404:6800:4000:1006::66

2404:6800:4000:1006::8a

2404:6800:4000:1006::71

2404:6800:4000:1006::65

142.250.134.139

142.250.134.100

142.250.134.113

142.250.134.102

142.250.134.101

142.250.134.138

4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

Ans :

200 : The request was successful and the server returned the requested data.

301 : The requested page has been permanently moved to a new URL.

404 : The requested page or resource does not exist on the server.

500 : The server encountered an unexpected problem while processing the request.

403 : The server understands the request but refuses to allow access.

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

Ans: **Root (.)**

- The highest level in DNS hierarchy.
- Represented by a dot (.) at the end of a domain name.
- Example: www.google.com.

2 TLD (Top-Level Domain)

- The extension of the domain name.
- In www.google.com, **.com** is the TLD.

3 Domain

- The registered name under the TLD.
- In www.google.com, **google** is the domain name.

4 Subdomain

- A subdivision of the main domain.
- In www.google.com, **www** is the subdomain.